

DEPRANIL TABLET 25MG

DESCRIPTION: A 7 mm round tablet, apple green in colour with 'duo' and 'd' markings.

COMPOSITION: Each tablet contains Clomipramine HCl 25 mg

PHARMACODYNAMICS: Pharmacotherapeutic group: Tricyclic antidepressant. Noradrenaline and preferential serotonin-reuptake inhibitor (ATC code N06A A04)

Mechanism of action: The therapeutic activity of Depranil is believed to be based on its ability to inhibit the neuronal re-uptake of noradrenaline (NA) and serotonin (5-HT) released in the synaptic cleft, with inhibition of 5-HT reuptake being the more important of these activities.

Depranil also has a wide pharmacological spectrum of action, which includes alpha1-adrenolytic, anticholinergic, antihistaminic, and antiserotonergic (5-HT-receptor blocking) properties.

Pharmacodynamic effects: Depranil acts on the depressive syndrome as a whole, including in particular typical features such as psychomotor retardation, depressed mood, and anxiety. The clinical response usually sets in after 2-3 weeks of treatment. Depranil also exerts a specific effect on obsessive-compulsive disorder distinct from its antidepressant effects. In chronic pain with or without somatic causes, Depranil acts presumably by facilitating serotonin and noradrenaline neurotransmission.

PHARMACOKINETICS: Absorption: Clomipramine is completely absorbed from the gastrointestinal tract. The systemic bioavailability of unchanged clomipramine is reduced to about 50% by hepatic first-pass metabolism to the active metabolite, *N*-desmethylclomipramine. The bioavailability of clomipramine is not markedly affected by the ingestion of food. Only the onset of absorption may be slightly delayed and therefore time to peak prolonged. During oral administration of constant daily doses of Depranil, the steady-state plasma concentrations of clomipramine show a high variability between patients. The dose of 75 mg daily, administered as coated tablets of 25 mg three times a day, produces steady-state plasma concentrations ranging from about 20 to 175 ng/mL. The steady-state plasma concentrations of the active metabolite *N*-desmethylclomipramine follow a similar pattern. However, at a dose of 75 mg Depranil per day, they are 40-85% higher than those of clomipramine.

Distribution: Clomipramine is 97.6% bound to plasma proteins. The apparent distribution volume is about 12 to 17 L/kg bodyweight. Concentrations in cerebrospinal fluid are about 2% of the plasma concentration. Clomipramine passes into maternal milk in concentrations similar to those in plasma.

Biotransformation: The primary route of clomipramine metabolism is demethylation to form the active metabolite, *N*-desmethylclomipramine. *N*-desmethylclomipramine can be formed by several P450 enzymes, primary CYP3A4, CYP2C19, and CYP1A2. Clomipramine and *N*-desmethylclomipramine are hydroxylated to form 8-hydroxyclopmipramine or 8-hydroxy-*N*-desmethylclomipramine. The activity of the 8-hydroxy metabolites are not defined in vivo. Clomipramine is also hydroxylated at the 2-position and *N*-desmethylclomipramine can be further demethylated to form didesmethylclomipramine. The 2- and 8-hydroxy metabolites are excreted primarily as glucuronides in the urine. Elimination of the active components, clomipramine and *N*-desmethylclomipramine, by formation of 2- and 8-hydroxy clomipramine is catalysed by CYP2D6.

Elimination: Clomipramine is eliminated from the blood with a mean half-life of 21 h (range: 12-36 h), and desmethylclomipramine with a mean half-life of 36 h.

About two thirds of a single dose of clomipramine are excreted in the form of water-soluble conjugates in the urine and approximately one third in the faeces. The quantity of unchanged clomipramine and desmethylclomipramine excreted in the urine is about 2% and 0.5% of the dose administered, respectively.

Characteristics in patients: In elderly patients, owing to reduced metabolic clearance, plasma clomipramine concentrations at any given dose are higher than in younger patients. The effects of hepatic and renal impairment on the pharmacokinetics of clomipramine have not been determined.

INDICATIONS: Treatment of depressive states of varying aetiology and symptomatology, e.g.

- endogenous, reactive, neurotic, organic, masked, and involutional forms of depression,
- depression associated with schizophrenia and personality disorders,
- depressive syndromes due to presenility or senility, to chronic painful conditions, and to chronic somatic diseases,
- depressive mood disorders of a reactive, neurotic, or psychopathic nature.

Obsessive-compulsive syndromes.

Phobias and panic attacks.

Cataplexy accompanying narcolepsy.

Chronic painful conditions.

Nocturnal enuresis (only in patients over the age of 5 and if organic causes have been excluded).

RECOMMENDED DOSAGE: This product is not able to deliver all approved dose regimens.

Before initiating treatment with Depranil, hypokalaemia should be treated (see Special warnings and special precautions for use).

The dosage and method of administration should be adapted to the individual patient's condition. The aim is to achieve an optimum effect while keeping the doses as low as possible and increasing them cautiously, particularly in elderly patients or adolescents, who generally show a stronger response to Depranil than patients of intermediate age groups.

As a precaution against possible serotonergic toxicity, adherence to the recommended doses of Depranil is advised and any increase in dose should be made with caution if other serotonergic agents are co-administered (see Special warnings and precautions for use and Interactions with other medicaments and other forms of interaction).

Depression, obsessive-compulsive syndromes, and phobias: Start treatment with 1 tablet of 25 mg 2-3 times daily. Raise the daily dosage stepwise, e.g. 25 mg every few days, (depending on how the medication is tolerated) to 4-6 tablets of 25 mg during the first week of treatment. In severe cases this dosage can be increased up to a maximum of 250 mg daily. Once there is a distinct improvement, adjust the daily dosage to a maintenance level of about 2-4 coated tablets of 25 mg.

Panic attacks, agoraphobia: Start with 1 tablet of 10 mg daily, possibly in combination with a benzodiazepine. Depending on how the medication is tolerated, raise the dosage until the desired response is obtained, while gradually withdrawing the benzodiazepine. The daily dosage required varies greatly from patient to patient and lies between 25 and 100 mg. If necessary it can be increased to 150 mg. It is advisable for treatment not to be discontinued for at least 6 months and for the maintenance dose to be reduced slowly during this time. This product is not able to deliver all approved dose regimens.

Cataplexy accompanying narcolepsy: Daily dose of 25-75 mg.

Chronic painful conditions: The dosage must be individualised [10-150 mg daily], while taking account of concomitant analgesic medication (and of the possibility of reducing use of analgesics).

Elderly patients: Start treatment with 1 tablet of 10 mg daily. Gradually raise the dosage to an optimum level of 30-50 mg daily, which should be reached after about 10 days and then maintained until the end of treatment. This product is not able to deliver all approved dose regimens.

Children and Adolescents: The starting dose is 25 mg daily and should be gradually increased (also given in divided doses) during the first 2 weeks, as tolerated, up to a daily maximum of 3 mg/kg or 100 mg, whichever is smaller. Thereafter, the dosage may be increased gradually over the next several weeks up to a daily maximum of 3 mg/kg or 200 mg, whichever is smaller.

Nocturnal enuresis: Initial daily dose in children aged 5-8 years, 2-3 coated tablets of 10 mg; children aged 9-12, 1-2 tablets of 25 mg; older children, 1-3 tablets of 25 mg. The higher doses are for patients who do not respond fully to treatment within one week. The coated tablets should normally be given in a single dose after the evening meal, but children who wet their beds early in the night should be given part of the dose beforehand (at 4 p.m.). Once the desired response has been achieved, treatment should be continued (for 1-3 months) and the dose gradually reduced. This product is not able to deliver all approved dose regimens.

No experience is available in children under 5 years.

CONTRAINDICATIONS: Hypersensitivity to clomipramine or any of the excipients, or cross-sensitivity to tricyclic antidepressants of the dibenzazepine group.

Depranil should not be given in combination, or within 14 days before or after treatment, with a MAO inhibitor (see Interactions with other medicaments and other forms of interaction). The concomitant treatment with selective, reversible MAO-A inhibitors, such as moclobemide, is also contraindicated.

Recent myocardial infarction.

Congenital long QT syndrome.

ROUTE OF ADMINISTRATION: Oral

WARNING & PRECAUTIONS:

Suicidality in Children and Adolescents

- Antidepressants increase the risk of suicidal thinking and behavior (suicidality) in children and adolescents with major depressive disorder (MDD) and other psychiatric disorders.
- Anyone considering the use of an antidepressant in a child or adolescent for any clinical use must balance the risk of increased suicidality with the clinical need.
- Patients who are started on therapy should be observed closely for clinical worsening, suicidality, or unusual changes in behavior.
- Families and caregivers should be advised to closely observe the patient and to communicate with the prescriber.
- Depranil is not approved for use in pediatric patients except for patients with obsessive compulsive disorder (OCD) and nocturnal enuresis.

Warnings: Tricyclic antidepressants are known to lower the convulsion threshold and Depranil should, therefore, be used with extreme caution in patients with epilepsy and other predisposing factors, e.g. brain damage of varying aetiology, concomitant use of neuroleptics, withdrawal from alcohol or drugs with anticonvulsive properties (e.g. benzodiazepines). It appears that the occurrence of seizures is dose dependent. Therefore, the recommended total daily dose of Depranil should not be exceeded.

Depranil should be administered with particular caution in patients with cardiovascular disorders, especially those with cardiovascular insufficiency, conduction disorders, (e.g. atrioventricular block grades I to III), or arrhythmias. Monitoring of cardiac function and the ECG is indicated in such patients, as well as in elderly patients.

There may be a risk of QTc prolongation and Torsades de Pointes, particularly at supra-therapeutic doses or supra-therapeutic plasma concentrations of clomipramine, as occur in the case of co-medication with selective serotonin reuptake inhibitors (SSRIs) or serotonin and noradrenergic reuptake inhibitors (SNaRIs). Therefore, concomitant administration of drugs that can cause accumulation of clomipramine should be avoided. Equally, concomitant administration of drugs that can prolong the QTc interval should be avoided (see Interactions with other medicaments and other forms of interaction). It is established that hypokalaemia is a risk-factor of QTc prolongation and Torsades de pointes. Therefore, hypokalaemia should be treated before initiating treatment with Depranil and Depranil should be used with caution when combined with SSRIs, SNaRIs or diuretics (see Interactions with other medicaments and other forms of interaction). Because of its anticholinergic properties, Depranil should be used with caution in patients with a history of increased intraocular pressure, narrow-angle glaucoma, or urinary retention (e.g. diseases of the prostate).

Due to the risk of serotonergic toxicity, it is advisable to adhere to recommended doses and increase the dose with caution if another serotonergic agent is co-administered. Serotonin Syndrome, with symptoms such as hyperpyrexia, myoclonus, agitation, seizures, delirium and coma, can possibly occur when clomipramine is administered with serotonergic co-medication such as SSRIs, SNaRIs, tricyclic antidepressants or lithium. For fluoxetine a washout period of two to three weeks is advised before and after treatment with fluoxetine.

Caution is called for when giving tricyclic antidepressants to patients with severe hepatic disease and tumours of the adrenal medulla (e.g. pheochromocytoma, neuroblastoma), in whom they may provoke hypertensive crises.

Many patients with panic disorder experience more marked anxiety at the start of the treatment with Depranil (see Dosage and method of administration). This paradoxical initial increase in anxiety is most pronounced during the first few days of treatment and generally subsides within two weeks.

Activation of psychosis has occasionally been observed in patients with schizophrenia receiving tricyclic antidepressants.

Hypomanic or manic episodes have also been reported during a depressive phase in patients with cyclic affective disorders receiving treatment with a tricyclic antidepressant.

In such cases it may be necessary to reduce the dosage of Depranil or to withdraw it and administer an antipsychotic agent. After such episodes have subsided, low dose therapy with Depranil may be resumed if required.

Precautions: Before starting treatment with Depranil, it is advisable to check blood pressure because patients with postural hypotension or a labile circulation may experience a fall in blood pressure.

Caution is indicated in patients with hyperthyroidism or patients receiving thyroid preparations, owing to the possibility of cardiac toxicity. In patients with liver disease, periodic monitoring of hepatic enzyme levels is recommended.

Although changes in the white blood cell count have been reported with Depranil only in isolated cases, periodic blood cell counts and monitoring for symptoms such as fever and sore throat are called for, particularly during the first few months of therapy and during prolonged treatment.

Like related tricyclic antidepressants, Depranil should be given with electroconvulsive therapy only under careful supervision.

In predisposed and elderly patients, tricyclic antidepressants may provoke pharmacogenic (delirious) psychoses, particularly at night. These disappear within a few days of withdrawing the drug.

Risk of suicide is inherent to severe depression and may persist until significant remission occurs. At the beginning of treatment, a combined therapy with benzodiazepines or neuroleptics may be indicated (see Special warnings and special precautions for use and Interactions with other medicaments and other forms of interaction). Depranil has been reported to be associated with fewer deaths following overdose than other tricyclic antidepressants.

Caution is called for in patients with chronic constipation. Tricyclic antidepressants may cause paralytic ileus, particularly in elderly and in bedridden patients.

Before general or local anaesthesia, the anaesthetist should be told that the patient has been receiving Depranil (see Interactions with other medicaments and other forms of interaction).

An increase in dental caries has been reported during long-term treatment with tricyclic antidepressants. Regular dental check-ups are therefore advisable during long-term treatment. Decreased lacrimation and accumulation of mucoid secretions due to the anticholinergic properties of tricyclic antidepressants may cause damage to the corneal epithelium in patients with contact lenses.

Abrupt withdrawal should be avoided because of possible adverse reactions (see Adverse effects).

INTERACTION WITH OTHER MEDICATIONS:

Pharmacodynamic-related interactions

Adrenergic neurone blockers^(*): Depranil may diminish or abolish the antihypertensive effects of guanethidine, betanidine, reserpine, clonidine and alpha-methyl dopa. Patients requiring comedication for hypertension should therefore be given antihypertensives of a different type (e.g. vasodilators, or beta-blockers).

Anticholinergic agents^(*): Tricyclic antidepressants may potentiate the effects of these drugs (e.g. phenothiazine, antiparkinsonian agents, antihistamines, atropine, biperiden) on the eye, central nervous system, bowel and bladder.

CNS depressants^(*): Tricyclic antidepressants may potentiate the effects of alcohol and other central depressant substances (e.g. barbiturates, benzodiazepines, or general anaesthetics).

Diuretics^(*): Diuretics may lead to hypokalaemia, which in turn increases the risk of QTc prolongation and Torsades de Pointes. Hypokalaemia should therefore be treated prior to administration of Depranil (see Dosage and Special warnings and precautions).

MAO inhibitors^(*): Do not give Depranil for at least 2 weeks after discontinuation of treatment with MAO inhibitors (there is a risk of severe symptoms such as hypertensive crisis, hyperpyrexia and those consistent with Serotonin Syndrome, e.g. myoclonus, agitation, seizures, delirium and coma). The same applies when giving a MAO inhibitor after previous treatment with Depranil. In both instances Depranil or the MAO inhibitor should initially be given in small, gradually increasing doses and its effects monitored.

There is evidence to suggest that Depranil may be given as little as 24 hours after a reversible MAO-A inhibitor such as moclobemide, but the two-week washout period must be observed if the MAO-A inhibitor is given after Depranil has been used.

Selective serotonin reuptake inhibitors (SSRI)^(*): Comedication with SSRIs may lead to additive effects on the serotonergic system (see serotonergic agents)

Serotonergic Agents: Serotonin Syndrome can possibly occur when clomipramine is administered with serotonergic co-medication such as selective serotonin reuptake inhibitors (SSRIs), serotonin and noradrenergic reuptake inhibitors (SNaRIs), tricyclic antidepressants or lithium. For fluoxetine, a washout period of two to three weeks is advised before and after treatment with fluoxetine.

Sympathomimetic drugs^(*): Depranil may potentiate the cardiovascular effects of adrenaline, noradrenaline, isoprenaline, ephedrine and phenylephrine (e.g. local anaesthetics).

^(*)subsections arranged alphabetically

Pharmacokinetic-related interactions

Depranil (clomipramine) is predominantly eliminated through metabolism. The primary route of metabolism is demethylation to form the active metabolite, *N*-desmethyldomipramine, followed by hydroxylation and further conjugation of both *N*-desmethyldomipramine and the parent drug. Several cytochrome P450s are involved in the demethylation, mainly CYP3A4, CYP2C19, and CYP1A2. Elimination of both active components is by hydroxylation and this is catalyzed by CYP2D6.

Concomitant administration of CYP2D6 inhibitors may lead to an increase in concentration of both active components, up to ~3-fold in patients with a debrisoquine/sparteine extensive metabolizer phenotype, converting them to a poor-metabolizer phenotype. Concomitant administration of CYP1A2, CYP2C19 and CYP3A4 inhibitors are expected to increase clomipramine concentrations and decrease *N*-desmethyldomipramine, thus not necessarily affecting the overall pharmacology.

MAO inhibitors, which are also potent CYP2D6 inhibitors *in vivo*, such as moclobemide, are contraindicated for coadministration with clomipramine.

Antiarrhythmics (such as quinidine and propafenone) which are potent inhibitors of CYP2D6, should not be used in combination with tricyclic antidepressants.

SSRIs which are inhibitors of CYP2D6, such as fluoxetine, paroxetine, or sertraline, and of others including CYP1A2 and CYP2C19 (e.g. fluvoxamine), may also increase plasma concentrations of clomipramine, with corresponding adverse effects. Steady-state serum levels of clomipramine increased ~4-fold by co-administration of fluvoxamine (*N*-desmethyldomipramine decreased ~2-fold).

Comedication of neuroleptics (e.g. phenothiazines) may result in increased plasma levels of tricyclic antidepressants, a lowered convulsion threshold, and seizures. Combination with thioridazine may produce severe cardiac arrhythmias.

Coadministration with the histamine₂ (H₂)-receptor antagonist, cimetidine (an inhibitor of several P450 enzymes, including CYP2D6 and CYP3A4), may increase plasma concentrations of tricyclic antidepressants, whose dosage should therefore be reduced.

No interaction between chronic oral contraceptive use (15 or 30 mg ethinyl estradiol daily) and Depranil (25 mg daily) has been documented. Estrogens are not known to be inhibitors of CYP2D6, the major enzyme involved in clomipramine clearance and, therefore, no interaction is expected. Although, in a few cases with high dose estrogen (50 mg daily) and the tricyclic antidepressant imipramine, increased side effects and therapeutic response were noted, it is unclear as to the relevance of these cases to clomipramine and lower dose estrogen regimens. Monitoring therapeutic response of tricyclic antidepressants at high dose estrogen regimens (50 mg daily) is recommended and dose adjustments may be necessary.

Methylphenidate (e.g. Ritalin) may also increase concentrations of tricyclic antidepressants by potentially inhibiting their metabolism and a dose reduction of the tricyclic antidepressant may be necessary.

Some tricyclic antidepressants may potentiate the anticoagulant effect of coumarin drugs, such as warfarin, and this may be through inhibition of their metabolism (CYP2C9). There is no evidence for the ability of clomipramine to inhibit the metabolism of anticoagulants, such as warfarin, however, careful monitoring of plasma prothrombin has been advised for this class of drug.

Concomitant administration of drugs known to induce cytochrome P450 enzymes, particularly CYP3A4, CYP2C19, and/or CYP1A2 may accelerate the metabolism and decrease the efficacy of Depranil.

CYP3A and CYP2C inducers, such as rifampicin or anticonvulsants (e.g. barbiturates, carbamazepine, phenobarbital and phenytoin), may decrease clomipramine concentrations. Known inducers

USE IN PREGNANCY & LACTATION: Experience with Depranil in pregnancy is limited. Since there have been isolated reports of a possible connection between the use of tricyclic antidepressants and adverse effects (developmental disorders) on the foetus, treatment with Depranil should be avoided during pregnancy, unless the anticipated benefits justify the potential risk to the foetus.

Since the active substance passes into the breast milk, Depranil should be gradually withdrawn or the infant weaned if the patient is breast-feeding.

Elderly patients are particularly sensitive to anticholinergic, neurological, psychiatric, or cardiovascular effects. Their ability to metabolise and eliminate drugs may be reduced, leading to a risk of elevated plasma concentrations at therapeutic doses.

Central nervous system

Common: confusion, disorientation, hallucinations (particularly in elderly patients and patients with Parkinson's disease), anxiety states, agitation, sleep disturbances, mania, hypomania, aggressiveness, impaired memory, depersonalisation, aggravated depression, impaired concentration, insomnia, nightmares, yawning.

The following symptoms commonly occur after abrupt withdrawal or reduction of the dose: nausea, vomiting, abdominal pain, diarrhoea, insomnia, headache, nervousness, and anxiety.

Treatment of symptoms is based on modern methods of intensive care, with continuous monitoring of cardiac function, blood gases, and electrolytes, and if necessary emergency measures such as anticonvulsive therapy, artificial respiration, and resuscitation. Since it has been reported that physostigmine may cause severe bradycardia, asystole, and seizures, its use is not recommended in cases of overdosage with Depranil. Haemodialyses or peritoneal dialyses are ineffective because of the low plasma concentrations of clomipramine.

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